

Positive linearization of finite-dimensional isotone contractions

The finite-dimensional case of Remark 4.9(c) in arXiv:1810.09684

Autonomous proof packet — agent_lane_12

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Abstract

Chill, Fiorenza, and Król ask whether exact interpolation for positive linear operators always implies exact interpolation for order-preserving Lipschitz operators with solid lattice domain. We prove the implication for every finite-dimensional compatible Banach-lattice couple. The key is a pairwise linearization theorem: for any two points f, h in the domain of an order-preserving endpoint contraction S , there is one positive admissible linear endpoint contraction T such that $T(f - h) = S(f) - S(h)$. We smooth S inside the order box generated by f, h ; its derivatives are positive endpoint contractions, and their average along the segment gives T in the limit. The argument includes proper solid domains, the feature not covered by the source's observation about everywhere-defined Lipschitz maps. The general infinite-dimensional question remains open.

1 Source problem and result

The source is R. Chill, A. Fiorenza, and S. Król, *Interpolation of nonlinear positive or order preserving operators on Banach lattices*, arXiv:1810.09684 [1]. Theorem 4.8 proves

$$\begin{aligned} \text{exact partially } K\text{-monotone} &\iff \text{exact positive Gagliardo–Peetre} \\ &\implies \text{exact order-preserving Lipschitz} \implies \text{exact positive linear.} \end{aligned}$$

Remark 4.9(c), on source PDF page 18, asks whether the last implication can always be reversed.

(c) We do not know if there are exact interpolation spaces for positive operators which are not exact interpolation spaces for order preserving Lipschitz operators, that is, more precisely, whether the implication (iv) $\Rightarrow$ (iii) is true in general or not.

An interpolation couple $\vec{E} = (E_0, E_1)$ is understood in the source's sense: E_0, E_1 , and $E_0 + E_1$ are Banach lattices, and the endpoint spaces are order ideals in the sum. For an admissible linear operator set

$$\|T\|_{\mathcal{L}(\vec{E})} = \max_{i=0,1} \|T : E_i \rightarrow E_i\|.$$

Theorem 1 (Finite-dimensional affirmative answer). *Let $\vec{E} = (E_0, E_1)$ be a compatible Banach-lattice couple for which $E := E_0 + E_1$ is finite-dimensional, and let X be an intermediate space. If X is an exact interpolation space for positive linear operators on $\vec{E}$, then it is an exact interpolation space for admissible order-preserving Lipschitz operators whose domains are solid lattices. Thus implication (iv) $\Rightarrow$ (iii) of source Theorem 4.8 holds for every finite-dimensional compatible Banach-lattice couple.*

The theorem follows immediately from the stronger pairwise statement below.

2 Pairwise positive linearization

Lemma 2. *Let $E = E_0 + E_1$ be finite-dimensional. Suppose that $S : D \rightarrow E$ is order preserving, $D \subset E$ is a solid vector lattice, and for some $L \geq 0$,*

$$\|Sx - Sy\|_{E_i} \leq L\|x - y\|_{E_i} \quad \text{whenever } x, y \in D \text{ and } x - y \in E_i, \quad i = 0, 1. \quad (1)$$

Then for every $f, h \in D$ there exists a positive admissible linear operator $T : E \rightarrow E$ such that

$$\|T\|_{\mathcal{L}(\vec{E})} \leq L, \quad T(f - h) = S(f) - S(h).$$

Proof. The case $f = h$ is immediate. A finite-dimensional vector lattice is lattice-isomorphic to coordinate-ordered $\mathbb{R}^m$, so we use that model. Put

$$a = f \wedge h, \quad b = f \vee h.$$

Since D is a lattice, $a, b \in D$; since it is solid, the complete order box $[a, b]$ lies in D . Let B be the coordinate band spanned by the active coordinates j for which $a_j < b_j$, and let P be its band projection. Every band projection is positive and contractive for every lattice norm. Moreover $P(f - h) = f - h$.

We first note that the restriction of S to the relative interior

$$U = \{x \in a + B : a_j < x_j < b_j \text{ on every active coordinate}\}$$

is locally Lipschitz in an ordinary finite-dimensional norm. Indeed, the order ideals E_0, E_1 are coordinate subspaces and their sum is E . Choose disjoint coordinate band projections R_0, R_1 with $R_0 + R_1 = I$ and $R_i E \subset E_i$. If x, y lie in an inner order box, then the coordinatewise mixed point $z = x + R_0(y - x)$ lies in the same box, and

$$S(y) - S(x) = [S(y) - S(z)] + [S(z) - S(x)].$$

The two differences are controlled by (1) in E_1 and E_0 , respectively. This proves the asserted local continuity and permits ordinary convolution inside U .

Let ρ_δ be a nonnegative smooth mollifier on B , supported in a ball of radius δ and of integral one. On an inner box define

$$S_\delta(x) = \int_B S(x - z) \rho_\delta(z) dz.$$

This map is smooth. It remains order preserving: whenever $x \leq y$ and all translates stay in U , each integrand in $S(y - z) - S(x - z)$ is positive. Consequently $DS_\delta(x) : B \rightarrow E$ is a positive linear map.

Convolution also preserves both endpoint Lipschitz estimates. If $v \in B \cap E_i$, then

$$\begin{aligned} \|S_\delta(x + tv) - S_\delta(x)\|_{E_i} &\leq \int_B \|S(x + tv - z) - S(x - z)\|_{E_i} \rho_\delta(z) dz \\ &\leq L|t| \|v\|_{E_i}. \end{aligned}$$

After division by $|t|$ and passage to the limit, the operator

$$A_x := DS_\delta(x)P : E \rightarrow E$$

is positive, admissible, and satisfies $\|A_x\|_{\mathcal{L}(\vec{E})} \leq L$.

Let $c = (a + b)/2$ and, for $0 < \varepsilon < 1$, set

$$f_\varepsilon = (1 - \varepsilon)f + \varepsilon c, \quad h_\varepsilon = (1 - \varepsilon)h + \varepsilon c.$$

These points and the segment between them lie in an inner box. Choose δ small enough for the preceding construction and average the derivatives along that segment:

$$T_{\varepsilon,\delta} = \int_0^1 DS_\delta(h_\varepsilon + t(f_\varepsilon - h_\varepsilon))P dt.$$

It is again positive and $\|T_{\varepsilon,\delta}\|_{\mathcal{L}(\vec{E})} \leq L$. The fundamental theorem of calculus gives

$$T_{\varepsilon,\delta}(f - h) = \frac{S_\delta(f_\varepsilon) - S_\delta(h_\varepsilon)}{1 - \varepsilon}. \quad (2)$$

First let $\delta \downarrow 0$ and then $\varepsilon \downarrow 0$. The local continuity established above shows that the right side of (2) converges to $S(f) - S(h)$. The family of positive admissible operators with endpoint norms at most L is bounded in the finite-dimensional space $\mathcal{L}(E)$: the fixed decomposition $E = R_0E + R_1E$ bounds its norm on all of E . It is also closed. Hence a subsequence of $T_{\varepsilon,\delta}$ converges to a positive admissible operator T with $\|T\|_{\mathcal{L}(\vec{E})} \leq L$, and (2) yields $T(f - h) = S(f) - S(h)$. $\square$

Proof of Theorem 1. Let S be an order-preserving admissible Lipschitz operator with solid lattice domain, let $L = \|S\|_{\text{Lip}}$, and choose f, h in its domain with $f - h \in X$. Lemma 2 supplies a positive admissible linear T satisfying $T(f - h) = S(f) - S(h)$ and $\|T\|_{\mathcal{L}(\vec{E})} \leq L$. Exactness of X for positive linear operators therefore gives

$$\|S(f) - S(h)\|_X = \|T(f - h)\|_X \leq L\|f - h\|_X,$$

which is precisely source assertion (iii). $\square$

3 Scope and upgrade audit

The proof covers arbitrary finite-dimensional compatible lattice norms and, crucially, arbitrary solid lattice domains. It is stronger than merely extending S : the linear operator may depend on the selected pair, which is all exact interpolation needs.

The finite-dimensional hypothesis enters twice. Convolution occurs on a Lebesgue-measured order box, and the bounded family of averaged derivatives is compact in $\mathcal{L}(E)$. For a general Banach lattice, restricting to the two-dimensional lattice spanned by $(f - h)^+$ and $(f - h)^-$ produces only a positive operator on that sublattice. Extending it to the whole couple while preserving both endpoint constants is exactly the positive-extension/band gluing property not available in the source's general setting. Endpoint Lipschitz assumptions also give no continuity in the $E_0 + E_1$ topology for differences lying in neither endpoint, so finite-band approximation does not close the gap. These are genuine obstructions to the present method, not claims that the full implication is false.

Bounded searches on 11 August 2026 covered the run's cheap indexes and arXiv/web queries by exact title, authors, Remark 4.9(c), implication (iv) $\Rightarrow$ (iii), and finite-dimensional order-preserving linearization. They found the source and related extension literature but no later resolution or matching finite-dimensional theorem. This is not a priority claim.

Human-review recommendation. Accept as a substantial partial result. Review chiefly the local two-endpoint continuity argument, preservation of endpoint constants under mollification, and the final compactness passage. Do not record the general infinite-dimensional problem as solved.

References

- [1] R. Chill, A. Fiorenza, and S. Król, *Interpolation of nonlinear positive or order preserving operators on Banach lattices*, arXiv:1810.09684 (2018); Positivity 24 (2020), 507–532, Theorem 4.8 and Remark 4.9(c).